

Savvy – Sprint 0 POC Results (Android)

POC 1: Focus and Commitment · POC 2: Card Unlock (NFC + QR) · POC 3: Parent Mode and Anti-Bypass

Result: POC 1 – **Go** | POC 2 – **Go with limitation** | POC 3 – **Go with limitation**

Tested on a Google Pixel 4 (Android 13) and a Samsung Galaxy M36 (Android 16) with representative scenarios. Testing on other brands (Xiaomi, OnePlus, Oppo, Vivo) belongs to the full development phase. Card taps were tested with the same system event a real tag sends; a physical tag test waits for your card samples. **“Alert only”** means we detect it and record it (or alert the parent), but Android does not let us stop it.

POC 1 – Focus and Commitment Engine

Area	What the app does	Limitation
Choosing apps	The user picks the distracting apps (Instagram and others) from a list of installed apps with names and icons. The choice is saved and comes back after Savvy is reinstalled.	Shows apps that appear on the home screen only, which avoids Google’s restricted “see all apps” permission. Hidden system apps are not listed.
App blocking	When a blocked app opens, a Savvy block screen replaces it. Two independent detectors run together: Accessibility (reacts in under 0.1 s) and Usage Access (reacts in about 0.3–0.6 s, measured 0.3 s on Samsung). Works when opened from the home screen, Recent apps and web links, when Savvy is closed, with no internet and in battery-saving deep sleep.	With Accessibility off, the app’s opening screen can flash for about half a second before blocking. Opening from a notification, Google Assistant, split screen and picture-in-picture still need a manual test. The website version of an app in a browser is not blocked (MVP blocks apps only). Clone apps and Android 15 “Private space” are still to test.
Focus modes and to-do	Study, Work and Sleep sessions. A to-do task can keep apps blocked for a set time or “until done”. When time is up the apps unlock on their own and the notification disappears. A live countdown shows in the notification.	Kept deliberately simple (title, length, status). Whether marking a task done also needs the card is a product decision.
6-hour and 24-hour commitment	Both work. The end time is kept on Savvy’s server and on the phone. Changing the phone’s clock does not end it early (tested +7 h on Pixel, +3 h on Samsung). It survives restart, app close and airplane mode, and comes back after Savvy is uninstalled and reinstalled.	After reinstall the user must grant permissions again and re-register the card. The rare case “clock changed + restarted + Savvy force-stopped” is not covered, but force stop is already a bypass in self-use (see POC 3).
Emergency exit	Always available from the block screen, limited (for example 2 per week), counted on the server and shown to the user. An optional waiting time works: the button shows “Available at hh:mm”, then “Confirm”.	Needs internet to check the allowance. The number per week and the waiting time are your product decision.
Insights	Daily screen time with minutes per app, focus time and streak. 35 days of history are kept on the phone.	Android keeps only a few days of detailed usage itself, so history starts from the day Savvy is installed.

POC 2 – Savvy Card Unlock (NFC + QR)

Area	What the app does	Limitation
NFC card unlock	The user holds the Savvy card to the phone while the block screen shows, and the apps unlock in about a third of a second. One action . If Savvy is closed, tapping the card opens Savvy and unlocks directly (no extra notification, unlike iPhone).	NFC must be switched on. Tested with the card’s system event, not yet with a physical card. Needs your vendor card samples before we can confirm.
Card checking	Each card is registered to one account. Another user’s Savvy card, a forged card and random NFC tags are refused. The user’s own signed card also unlocks offline and is synced to the server later.	A standard NFC card can be copied with the right tools. A secure card (NXP NTAG 424 DNA) cannot be usefully copied, but costs more and needs internet to check.
QR fallback	The QR printed on the card holds the same card link as the chip, so one card works for NFC and QR on both iPhone and Android. Scanned with Savvy’s camera screen or the phone camera. Importing a QR from Photos is not offered.	A QR can be photographed, so it proves which card it is, not that the card is present. We recommend QR unlocks work after a short wait, or are limited like emergency exits. Camera scan on a phone still needs a manual test.

POC 3 – Parent Mode and Anti-Bypass

Area	What the app does	Limitation
Parent controls	The parent links the child's phone with a code and sees the child's apps with real names. The parent picks always-blocked apps, the focus mode, the unlock rule (card / locked / free) and a length of up to 24 h. On the child's phone the account, role and app list are locked, and leaving parent mode needs the parent's PIN.	No instant push yet: a new rule reaches the child within 15 minutes (instant delivery needs Firebase). One child per parent in the MVP. In production the PIN should be checked on the server.
Uninstall protection (parent mode)	With device administrator on, Savvy cannot be uninstalled, and "Force stop" and "Clear data" are disabled. The screens used to turn Savvy off (device admin, Accessibility, Usage Access, uninstall) are closed within 0.03–0.13 s. The parent sees when protection is off.	Alert only: Safe Mode, factory reset, and a child using USB debugging to clear Savvy's data (the parent sees "USB debugging on"). Android 17 Advanced Protection can turn off Accessibility; Usage Access keeps blocking.
Commitment lock (self-use)	Restart, closing Savvy, airplane mode and clock changes do not end a commitment. Turning off one permission leaves the other detector blocking. Uninstalling and reinstalling brings the commitment back. Savvy records and alerts when Accessibility is turned off.	Not possible on Google Play: blocking a self-use user from uninstalling Savvy or turning off its permissions. Alert only: uninstall, force stop and clear data (blocking resumes when Savvy is opened again), Safe Mode.
Phone brands	Savvy opens each brand's "keep running" screen (Samsung, Xiaomi, Oppo, Vivo, OnePlus, Huawei). A background job every 15 minutes restarts protection if the brand closed it. Samsung's screen tested and working.	Savvy cannot read these brand settings back, so it cannot confirm the user changed them. Xiaomi, OnePlus, Oppo and Vivo still need testing.

Decision needed – Device Owner mode. If Savvy is set up during a factory reset, most "Alert only" items become **fully prevented**: blocked apps are refused by Android itself with no flash, uninstall is blocked and Safe Mode is disabled. This works in our POC. The cost is that the phone must be reset, and it is not a Google Play consumer feature. **Our recommendation:** launch as a normal Play Store app and offer Device Owner later as an optional "maximum protection" mode for parents.

Other decisions needed from you

1. **QR rule:** instant, after a short wait, or limited per week.
2. **Card type:** standard NFC card (low cost, copyable) or secure NTAG 424 DNA card (cannot be usefully copied, needs internet).
3. **Emergency exit:** how many per week, and whether there is a waiting time.
4. **To-do tasks:** does marking a task done need the card?
5. **Parent mode:** can the child's own card unlock the parent's rules, or only the parent?
6. **Wording:** we recommend "commitment lock" instead of "uninstall protection" for self-use.

Permissions Needed

Permission	Used for
Accessibility (Settings)	Fast blocking of chosen apps; in parent mode only, guarding the screens used to remove Savvy
Usage Access (Settings)	Second blocking detector; screen-time summary
Display over other apps	Showing the block screen over the blocked app
Foreground service	Keeping the session running, with the countdown notification
NFC and Camera	Reading the Savvy card and scanning its QR
Notifications, Battery, brand settings	Session countdown, alerts, staying active
Device admin (parent mode only)	Blocking uninstall, force stop and clear data on the child's phone

Accessibility, Usage Access and the foreground service are sensitive: Google Play requires a declaration, an in-app disclosure and a demo video for each. Using Accessibility to stop removal is allowed only in parent-authorized parental control, so Savvy enables it in parent mode only. Savvy does not use the restricted "see all apps" permission. Approval is known only after submission.